AURA

ECLSS PREDICTIVE MAINTENANCE SYSTEM

User Manual

Complete operational guide covering installation, navigation, all system modules, ML/AI subsystems, settings, and troubleshooting.

FastAPI Backend | PyTorch · LSTM · DQN | scikit-learn · IF · RF | Chart.js · Three.js | SQLite WAL | Ollama · Groq AI

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1 · Introduction & Overview

1.1 What is AURA?

AURA (Autonomous Understanding and Response Architecture) is a real-time predictive maintenance platform purpose-built for the International Space Station's Environmental Control and Life Support System (ECLSS). It combines a physics-based sensor data generator, a four-layer machine learning pipeline, interactive 3D visualization, and a natural-language AI analyst into a single browser-accessible application.

AURA monitors seven ISS modules continuously, applies anomaly detection and fault classification to every reading, predicts failure probability and remaining useful life, and recommends remediation actions — all in real time, updating every second.

1.2 System Capabilities

⬡

3D Digital Twin

Interactive Three.js model of the ISS. Modules pulse red on anomaly. Click any module to jump to its sensor detail.

◈

4-Layer ML Pipeline

Isolation Forest → Random Forest → LSTM predictor → DQN recommender runs on every tick.

↻

Live Sensor Data

Physics-correlated data generator simulates 20 ECLSS parameters per location with configurable noise and fault drift.

◐

Trend Detection

Mann-Kendall, CUSUM, and z-score analysis detect slow degradation before IF anomaly threshold is breached.

✦

AI Analyst

Chat with a local Ollama or cloud Groq LLM. Full system context injected automatically. Persistent chat memory.

⚙

Secured Settings

All thresholds, generation parameters, and integrations are hot-reloadable without server restart. bcrypt auth.

1.3 Architecture Overview

LAYER	TECHNOLOGY	ROLE
Web Server	FastAPI + Uvicorn	REST API, WebSocket live feed, static file serving
Database	SQLite (WAL mode)	Sensor readings, anomaly labels, alerts, fault state
Data Generation	NumPy + Cholesky noise	Physics-correlated sensor simulation with fault drift
Anomaly Detection	Isolation Forest (sklearn)	Flags statistically anomalous readings each tick
Fault Classification	Random Forest (sklearn)	Classifies fault type and confidence from sensor pattern
Failure Prediction	LSTM + Multi-head Attention (PyTorch)	Rolling buffer → failure probability + RUL hours
Remediation	DQN (PyTorch)	Recommends optimal corrective action per location state

LAYER	TECHNOLOGY	ROLE
Trend Analysis	Mann-Kendall / CUSUM / Z-score (NumPy)	Detects slow degradation trends before acute anomaly
AI Chat	Ollama (local) / Groq API (cloud)	NL question answering with full system context injection
Frontend	Vanilla JS + Chart.js + Three.js	Single-page application, no build step required

> **ISS Locations Monitored:** JLP & JPM · Node 2 · Columbus · US Lab · Cupola · Node 1 · Joint Airlock

2 · Installation & Setup

2.1 System Requirements

REQUIREMENT	MINIMUM	NOTES
Python	3.10+	3.12 recommended. Must be accessible as `python3`
RAM	4 GB	8 GB recommended when LSTM + DQN models are loaded
Disk	500 MB	Includes Python venv, models, and SQLite database
OS	Linux or Windows 10+	macOS untested but should work via the Linux script
Browser	Chrome 93+ / Firefox 92+	WebSocket + ES2020 required. Safari not tested
Network	Local only	All traffic stays on localhost:8000 by default
Ollama (optional)	Latest	For local AI Analyst. `ollama pull mistral`

All Python dependencies are listed in `requirements.txt` and installed automatically by the start scripts:

```
fastapi==0.115.0    uvicorn[standard]==0.30.6    pydantic==2.9.2
numpy==2.1.3        scikit-learn>=1.5.2          joblib==1.4.2
ollama==0.3.3       groq>=0.9.0                  torch>=2.0.0
bcrypt>=4.0.0       python-jose[cryptography]>=3.3.0
cryptography>=41.0.0
```

2.2 Starting on Linux

1. Open a terminal and navigate to the project root directory.
2. Make the start script executable if needed: `chmod +x run/linux/start.sh`
3. Run the script: `bash run/linux/start.sh`
4. On the **first run**, the script creates a Python virtual environment (`.venv/`) and installs all dependencies. This may take 2–5 minutes depending on internet speed and whether PyTorch needs to download.
5. Subsequent runs detect if `requirements.txt` is unchanged (via SHA-256 hash stored in `.req_hash`) and skip reinstallation — startup takes only 2–4 seconds.
6. Open **http://localhost:8000** in your browser.

> **Dependency change detection:** If you ever manually add packages to `requirements.txt`, delete `run/linux/.req_hash` to force a fresh `pip install` on next launch.

2.3 Starting on Windows

1. Double-click `run\windows\start.bat` or run it from a Command Prompt.

2. The script creates `.venv\` and installs dependencies on the first run (same hash-based skip mechanism as Linux).
3. A browser window opens automatically to **http://localhost:8000**.

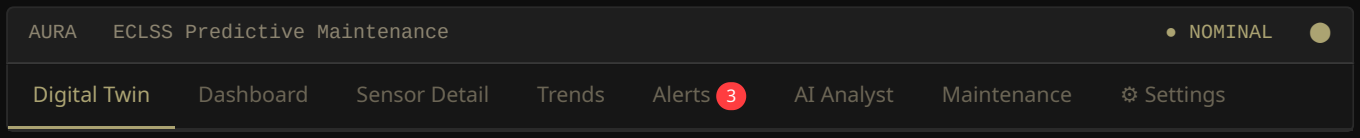
2.4 First-Run Checklist

- The database `data/aura.db` is created automatically if it does not exist.
- ML model files are expected in the `models/` directory. If missing, AURA still starts but the ML pipeline runs in a fallback mode (anomaly detection disabled).
- The default admin password for Settings is `admin`. Change it immediately via Settings → Security.
- For AI Analyst: ensure Ollama is running (`ollama serve`) and the desired model is pulled (`ollama pull mistral`). Alternatively, configure a Groq API key in Settings → Integrations.
- The encryption key for the Groq API key is stored in `.fernet.key` in the project root. This file is machine-local and should never be committed to version control.

> **Security note:** AURA is designed for local or trusted-network use. The web server binds to `0.0.0.0:8000` by default. Do not expose port 8000 to the public internet without adding reverse-proxy authentication.

3 · Interface Navigation

AURA is a single-page application. All navigation happens through the tab bar at the top of the screen — no page reloads occur and the WebSocket connection is maintained throughout.



TAB	PURPOSE	KEY INTERACTION
Digital Twin	3D ISS model with live module health indicators	Click a module → jumps to Sensor Detail for that location
Dashboard	Card-per-location overview with ML status	Live DQN action + LSTM status badge on every card
Sensor Detail	Mini sparkline grid → full drilldown chart	Click any sparkline card → 100–1000 point drilldown
Trends	Statistical trend analysis per location	Select location, click Analyze (auto-refreshes every 30s)
Alerts	Alert history with severity and acknowledgement	Filter by location, acknowledge individually or bulk
AI Analyst	LLM chat with full system context	Type questions; history persists across page refreshes
Maintenance	MTBF-based replacement and calibration schedules	Real-time drift accumulation and urgency status
⚙ Settings	All system configuration behind bcrypt auth	Login required; all changes hot-reload without restart

Status Indicators

The top-right corner shows two indicators:

- **System Status badge** — `• NOMINAL` when all locations are healthy; transitions to `• ANOMALY` if any location has an active anomaly.
- **WebSocket indicator** — green dot = live connection; red/grey = disconnected (auto-reconnects after 3 seconds).

The **Alerts** tab shows a red badge with the count of unacknowledged alerts. This count refreshes every 30 seconds as a fallback in case a WebSocket alert message is missed.

4 · Digital Twin

The Digital Twin page renders a real-time Three.js 3D model of the International Space Station. Each module is colored and animated to reflect its current health state as reported by the ML pipeline.

Module Health Visualization

MODULE STATE	VISUAL	MEANING
NOMINAL	Default module color, static	All sensors within normal range; no active fault
ANOMALOUS	Module pulses red	Isolation Forest flagged the latest reading as anomalous
FAULT LATCHED	Module glows amber	Random Forest has confirmed a specific fault with high confidence; fault is pinned until manually resolved

Camera Controls

- **Left-click + drag** — orbit the model (rotate view)
- **Scroll wheel** — zoom in/out
- **Right-click + drag** — pan
- **Reset Camera button** — returns to default isometric view (bottom-right of canvas)

Module Status Sidebar

The right sidebar lists all seven ISS locations with their current status. Each entry shows the location name, whether it is anomalous, and the active injected fault name (if any). The sidebar updates in real time from WebSocket tick messages.

Clicking a Module

Clicking any module in the 3D view automatically selects that location in the Sensor Detail tab's location dropdown and navigates to the Sensor Detail page, showing the overview grid for that module.

> **Fault Confirmation Popup:** When the Random Forest classifier detects the same fault class at ≥95% confidence for 3 or more consecutive ticks, a modal alert pops up with the fault name, location, and confidence. Click **Acknowledge** to dismiss. The fault remains latched (visible in Dashboard and Digital Twin) until explicitly resolved via Settings → Faults → Resolve.

5 · Dashboard

The Dashboard displays a card for each of the seven ISS modules, showing a comprehensive snapshot of each module's health, current readings, ML pipeline status, and recommended action. Cards update in real time from WebSocket tick messages.

Location Card Anatomy

US Lab • NOMINAL

O_2 partial pressure 21.2% | CO_2 partial pressure 0.38% | Temperature 22.1°C

Cabin pressure 14.7 psi | Humidity 45% | Airflow rate 0.55 m/s

DQN: **No Action Needed** | LSTM: **NOMINAL**

CARD ELEMENT	DESCRIPTION
Location name + status badge	NOMINAL or ANOMALOUS based on latest Isolation Forest label
Sensor readings grid	Live values for all 20 ECLSS parameters for that location. Values outside nominal range are highlighted.
Latched fault banner	Appears when the RF classifier has confirmed a fault. Shows fault name and confidence %.
DQN action	Current recommended remediation action from the Deep Q-Network. Updates every tick.
LSTM status	One-word failure probability status: NOMINAL / ELEVATED / WARNING / CRITICAL. Derived from LSTM's failure_prob output.

LSTM Status Words

STATUS	FAILURE PROBABILITY RANGE	MEANING
NOMINAL	< 20%	No imminent failure detected
ELEVATED	20%–49%	Subtle signal; monitor closely
WARNING	50%–79%	Significant precursor signal; investigate
CRITICAL	≥ 80%	High probability of imminent failure; take action

Refresh Behavior

Cards are updated continuously via the WebSocket feed. The **Refresh** button in the dashboard header triggers a full re-fetch of all location states from the REST API, useful if the WebSocket was temporarily disconnected.

6 · Sensor Detail

Sensor Detail provides a two-mode view for deep inspection of individual sensor parameters. Select a location from the dropdown at the top to load its sensor overview grid.

6.1 Overview Grid

The overview grid shows a mini sparkline card for every sensor parameter at the selected location. Each card displays:

- **Parameter name** — e.g., "O_2 partial pressure"
- **Current value** with unit
- **Mini sparkline chart** — last 50 readings, color-coded: normal points are gold (`#aba372`), anomalous points are red (`#ff3d40`)
- **Anomaly indicator** — card border pulses red if the latest reading is anomalous

Live WebSocket ticks push new data points to all visible sparklines in real time, so the overview is always current without requiring manual refresh.

6.2 Drilldown View

Click any mini sparkline card to enter drilldown mode for that parameter. The overview grid is replaced by a full-resolution chart and a data table.

Drilldown Controls

CONTROL	FUNCTION
← **Overview** button	Returns to the overview grid
Points dropdown	Switch between 100 / 200 / 400 / 1000 data points
Reload button	Fetches the selected number of historical points from the database
Live checkbox	When checked, new WebSocket ticks push the latest reading to the right of the chart in real time (oldest point drops off the left)

Anomaly Highlighting

Data points confirmed as anomalous by the Isolation Forest are rendered as **red dots** on the chart and as **red-highlighted rows** in the data table below the chart. Non-anomalous points are rendered in gold. This coloring uses a callback function per point so it cannot drift or shift as new live data arrives.

Data Table

Below the chart, a scrollable table lists every plotted reading with its timestamp and value. Anomalous rows are highlighted in red. The table is sorted oldest-to-newest (top to bottom).

7 · Trends Analysis

The Trends page applies multi-method statistical analysis to detect gradual degradation trends that may not yet register as anomalies in the Isolation Forest. This gives earlier warning of slow-developing faults (e.g., filter saturation, slow coolant leaks).

Controls

- **Location** — select which ISS module to analyze
- **Readings** — number of most recent readings to analyze (60 / 100 / 200)
- **Analyze** button — runs the analysis immediately
- **Auto-refresh (30s)** checkbox — automatically re-runs analysis every 30 seconds

7.1 Statistical Methods

Every parameter is evaluated against four independent tests simultaneously:

`Mann-Kendall Trend Test`
: Non-parametric test for monotonic trend. Computes τ (tau) and p-value. A statistically significant trend ($p <$ threshold, $|\tau| >$ advisory threshold) triggers at minimum an Advisory. Higher tau magnitude combined with direction toward a safety boundary escalates to Warning.

`Sen's Slope`
: Estimates the rate of change per reading in physical units. Acts as a magnitude gate — a statistically significant trend must also show meaningful rate of change (configurable slope gate) to avoid advisories on imperceptibly slow drift.

`CUSUM (Cumulative Sum)`
: Detects persistent upward or downward shifts by accumulating deviations from a baseline established from the first 15–40 readings. A CUSUM value exceeding the threshold indicates a sustained mean shift. Contributes one "signal" toward multi-signal confirmation.

`Rolling Z-Score`
: Computes the z-score of the most recent N readings relative to a rolling mean/std window. High z-scores indicate recent values are statistically unusual relative to recent history. Two thresholds: standard (requires 2+ signals) and single-signal (very high z, fires alone).

7.2 Severity Levels

SEVERITY	CONDITION	INTERPRETATION
NOMINAL	No test signals fire	Parameter is stable
ADVISORY	MK significant ($p < 0.01$, $\|\tau\| > 0.35$) + slope gate + ≥2 total signals	Slow trend detected; monitor
WARNING	Advisory conditions met AND trend direction is toward a safety boundary	Parameter drifting toward out-of-spec; investigate
CRITICAL	MK highly significant ($\|\tau\| > 0.65$) + all signals + rapid slope	Imminent boundary breach; act now

Each trend card also displays the **trend direction** (increasing / decreasing / no trend), the **current value** with the nominal range, and the **Sen's slope** rate of change.

> **Multi-signal requirement:** AURA requires at least two independent test signals to fire before issuing an Advisory. A single elevated z-score reading alone will not trigger an advisory unless it exceeds the single-signal threshold (default 4.5σ). This prevents false alerts from isolated noise spikes.

8 · Alerts

The Alerts page shows the historical log of all anomaly alert events. An alert is fired when the Isolation Forest flags a location as anomalous for a configurable number of consecutive ticks (default 10), subject to a cooldown period between repeated alerts for the same location.

Alert Firing Logic

1. Isolation Forest flags a reading as anomalous (label = -1).
2. The consecutive anomaly counter for that location increments.
3. When the counter reaches **Min Consecutive** (default 10), check: has the cooldown period elapsed since the last alert for this location?
4. If yes, fire the alert. Record severity based on Random Forest confidence: ≥ 85% RF confidence → CRITICAL; otherwise WARNING.
5. The counter resets on any nominal reading.

Alert Severity

SEVERITY	CONDITION	COLOR
WARNING	Sustained anomaly detected; RF confidence below CRITICAL gate	Amber
CRITICAL	Sustained anomaly + RF fault classification ≥ 85% confidence	Red

Alert Card Information

Each alert card displays:

- **Timestamp** — when the alert was fired
- **Location** — which ISS module triggered the alert
- **Severity** — WARNING or CRITICAL
- **Fault type** — the RF-classified fault name (if confidence was sufficient)
- **RF confidence** — probability score of the fault classification
- **Sensor snapshot** — the sensor readings at the time of the alert
- **Acknowledge button** — marks the alert as reviewed

Filtering

- **Unacknowledged only** checkbox — hide alerts that have already been acknowledged
- **Location filter** — limit the list to a specific module
- **Acknowledge All** — bulk-acknowledge all visible alerts

Alert badge: The red badge on the Alerts tab in the navigation bar shows the current count of unacknowledged alerts. It refreshes from the WebSocket on every new alert and polls the REST API every 30 seconds as a fallback.

9 · AI Analyst

The AI Analyst is an LLM-powered chat interface that has read-only access to all live ECLSS data. Before sending each message, AURA injects a full system context block including current sensor readings, anomaly states, active faults, recent alerts, and ML model outputs for all seven locations.

Sidebar Controls

CONTROL	FUNCTION
Scope dropdown	Focus the AI's context on a specific location or "All Locations"
Model dropdown	Select which Ollama model to use (mistral, llama3.1:8b, llama3)
Backend badge	Shows whether Ollama (local) or Groq (cloud) is responding, with connection status
Clear Chat	Clears both the visible messages and the browser localStorage history for this session
Live Anomalies	Live list of any currently anomalous parameters across all locations, updated per tick

9.1 Ollama (Local AI)

When Ollama is running locally (`ollama serve`), AURA connects to it at `http://localhost:11434`. Responses are streamed token-by-token and displayed in real time. Recommended models: **mistral** (fast, general purpose), **llama3.1:8b** (higher quality), **llama3** (balanced).

Ollama is fully private — no data leaves your machine. Response latency depends on hardware (GPU recommended for models ≥ 7B parameters).

9.2 Groq (Cloud Fallback)

If Ollama is unavailable, AURA automatically falls back to the Groq API (if a key is configured in Settings → Integrations). Groq serves open models (Mixtral, LLaMA) on dedicated hardware with very low latency. Responses are also streamed.

Configure the Groq API key in Settings → Integrations → Groq API. The key is stored encrypted on disk using Fernet symmetric encryption.

9.3 Chat Memory

Chat history is persisted to browser **localStorage** so conversations survive page refreshes and tab closes. The AI receives the last N messages (configurable in Settings → Display → Max chat stored) as conversation history, enabling follow-up questions without re-stating context.

The **Clear Chat** button wipes both the visible chat and the stored localStorage history.

Example Questions

- "Why is Node 2 showing anomalous readings?"
- "What is the current CO_2 trend across all modules?"
- "The LSTM is showing WARNING for US Lab — what should I check?"

- "Summarize all active alerts and recommend priority actions."
- "What is the RUL estimate for Joint Airlock and what does the DQN recommend?"

> **Context injection:** Every message sent to the AI includes a full system state block. This means the AI always has current data — you do not need to paste readings or describe conditions manually. Simply ask in natural language.

10 · Maintenance Schedule

The Maintenance page provides two evidence-based schedules: a **subsystem replacement schedule** based on observed ISS ECLSS MTBF data, and a **sensor calibration schedule** tracking accumulated drift against calibration thresholds.

Mission Clock

The mission elapsed time counter in the page header shows how long AURA has been running since the configured mission start date (set in Settings → Display → Mission Start). This drives the MTBF life-fraction calculations.

Subsystem Replacement Schedule

Each subsystem card shows:

- **Subsystem name** and related fault type
- **MTBF** — mean time between failures in hours, sourced from NASA post-flight anomaly reports
- **% Life Used** — mission elapsed hours ÷ MTBF
- **Hours Remaining** to next expected service interval
- **Maintenance type** — calendar-based (replace on schedule) or condition-based (replace when threshold breached)
- **Status badge** — NOMINAL / DUE SOON / OVERDUE based on % life used

MTBF values are calibrated to actual ISS operational experience (e.g., OGS: 3,104 h actual vs. 8,437 h pre-flight estimate).

Sensor Calibration Schedule

The calibration table tracks drift accumulation for sensors with known drift rates (sourced from ICES papers and JSC-62802). Columns:

COLUMN	DESCRIPTION
Sensor / Parameter	Name of the sensor and its parameter
Drift Rate	Fraction of nominal span per week (from calibration literature)
Accumulated Drift	Total drift since last calibration baseline, as % of span
Weeks Until Due	Weeks remaining before accumulated drift exceeds the 2% threshold
Status	OK / DUE SOON / CALIBRATE NOW

Note on TCCS: The Trace Contaminant Control System charcoal bed life is ~19,746 hours (2.25 years). There were no TCCS flight failures in 20+ years of ISS operations. A 90-day sampling cycle does NOT mean 90-day replacement — this is a common misconception. AURA uses the corrected MTBF value.

11 · Settings

All system configuration is accessible via the Settings tab. Settings are protected by a bcrypt-hashed admin password. All changes hot-reload into the running server — no restart is required. Settings are persisted to `settings.json` in the project root.

11.1 Authentication

When navigating to Settings, a login wall is displayed. Enter the admin password and click **Unlock**. A JWT token (30-minute TTL) is issued and stored in `sessionStorage` — it is automatically cleared when the browser tab is closed.

- **Default password:** `admin` — change immediately via Security tab
- **Lock button** — manually invalidates the session token and returns to login wall
- The token is sent as a `Bearer` header on every Settings API request
- If the server is restarted, all tokens are invalidated (new JWT secret generated at startup)

11.2 Data Management

Destructive data operations, now with watermarked selective export:

ACTION	EFFECT
Clear Sensor Data	Deletes all rows from `generated_data` and `anomaly_labels`; resets LSTM rolling buffers
Clear Alert History	Deletes all alert records from the `alerts` table
Clear Active Faults	Clears all injected faults and resets latch/streak counters
Clear LSTM Buffers	Clears LSTM rolling windows; use after sensor recalibration events
Export to CSV	Downloads all sensor data as a CSV file; records a watermark ID for selective clearing
Clear Exported Data	Deletes only rows with ID ≤ watermark (set at export time); rows generated after export began are safe

CSV Export Format

Exported CSV columns:

```
id, timestamp, location, parameter, value, is_anomalous, rf_prediction, rf_confidence
```

One row per sensor parameter per tick. A single tick for all 7 locations with 20 parameters each yields up to 140 CSV rows per second of simulation time.

11.3 Alert Thresholds

SETTING	DEFAULT	DESCRIPTION
Min consecutive anomalous ticks	10	How many consecutive anomalous IF readings before an alert fires. Higher = fewer false positives.
Alert cooldown (seconds)	600	Minimum time between repeated alerts for the same location. Prevents alert flooding.
CRITICAL RF confidence gate	0.85	RF confidence must exceed this threshold for an alert to be classified CRITICAL (vs. WARNING).
Latch RF confidence threshold	0.95	RF confidence must exceed this for a fault classification to "latch" (pin the fault as confirmed).
Latch min consecutive ticks	3	Number of consecutive high-confidence ticks required before latching a fault.
DQN RF bypass threshold	0.85	When RF confidence exceeds this, the DQN uses the RF-identified fault directly rather than the sensor state vector.

11.4 Data Generation

SETTING	DEFAULT	DESCRIPTION
Tick interval (seconds)	1.0	How often the data generation loop runs. Lower = more data but higher CPU usage.
Noise scale multiplier	1.0	Scales all per-sensor noise sigma values. 2.0 = twice the noise; 0.5 = half.
Crew event frequency	medium	How often transient crew-activity events occur (breathing, exercise, etc.). Affects O_2/CO_2/humidity.
Fault injection enabled	true	Master switch for fault drift. When off, injected faults have no effect on sensor readings.

11.5 Fault Management

The Faults tab replaces the old twin sidebar controls with a more comprehensive interface:

- **Active Faults table** — shows the current injected fault (if any) for each of the 7 locations, with a per-row **Resolve** button to clear that location's fault
- **Inject Fault form** — select a location and fault type; a description panel shows the fault's precursor detection window and which sensor parameters it affects; click **Inject Fault** to apply
- **Clear All Injected Faults** — removes all injected faults at once

> **Important:** Injecting a fault only starts the drift — the ML pipeline may take several ticks to detect it depending on the fault type and current thresholds. Faults with fast drift rates (e.g., NH_3 Coolant Leak, O_2 Leak) produce anomaly signals within 1–3 minutes. Slow faults (e.g., Water Processor Failure, Trace Contaminant Filter Saturation) may take much longer.

11.6 ML Models

The ML Models tab shows the load status of all four ML components. No configuration is available — this is a diagnostic view only.

COMPONENT	STATUS FIELD	NOTES
Isolation Forest + Random Forest	Loaded / Not loaded	sklearn version displayed; retrain recommended if version mismatch
LSTM Predictor	Loaded / Not loaded	PyTorch model; requires sufficient data in rolling buffer before producing predictions
DQN Recommender	Loaded / Not loaded	Deep Q-Network; falls back to rule-based recommendations if not loaded

11.7 Trend Detection Sensitivity

All trend thresholds can be adjusted without restarting the server:

SETTING	DEFAULT	EFFECT OF INCREASING
Mann-Kendall p-value threshold	0.01	Stricter statistical significance required (fewer advisories)
MK τ advisory threshold	0.35	Stronger trend required for advisory (fewer advisories)
MK τ warning threshold	0.65	Stronger trend required for warning (fewer warnings)
Slope magnitude gate	0.05	Larger rate of change required to confirm significance (fewer alerts)
CUSUM threshold	7.0	Higher cumulative deviation needed before CUSUM signals
CUSUM baseline window	0.20	Larger fraction of data used as baseline (more stable baseline)
Z-score threshold (2-signal)	3.5	Higher z-score required for contributing advisory signal
Z-score single-signal threshold	4.5	Higher z-score required to fire alert on its own
Rolling z-score window	30	Larger window = smoother baseline = less sensitive to recent changes

11.8 Integrations

Groq API

Groq provides cloud-hosted inference for open LLMs (Mixtral, LLaMA) with very low latency. AURA uses Groq as a fallback when Ollama is unavailable.

1. Create a free account at console.groq.com and generate an API key (format: `gsk_…`).
2. In Settings → Integrations → Groq API, paste your key and click **Save Key** .
3. Click **Test Connection** to verify the key is valid.
4. The key is stored encrypted on disk using Fernet symmetric encryption. It is never stored in plaintext.

11.9 Display & UI

SETTING	DEFAULT	DESCRIPTION
Mission start datetime	(empty)	ISO-format start time for the mission clock (e.g., `2025-01-01T00:00:00`)
Dashboard auto-refresh (ms)	5000	How often the dashboard polls for updates (in addition to WebSocket push)
Max chat messages stored	50	Maximum chat messages kept in localStorage for AI context
Default trends N readings	100	Pre-selected reading count when opening the Trends page
Default detail chart N points	100	Pre-selected point count when opening a sensor drilldown

11.10 Security

Change Admin Password

Enter the current password, then the new password twice (minimum 6 characters). Passwords are hashed with bcrypt before storage. The plaintext password is never stored or logged.

Revoke All Sessions

Clicking **Revoke All Sessions** generates a new JWT secret on the server. This immediately invalidates every active session token including the current one. You will be automatically logged out and returned to the login wall. Use this if you suspect an unauthorized session is active.

> **Session storage:** Tokens are stored in `sessionStorage`, not `localStorage`. They are automatically cleared when the browser tab is closed. There is no "remember me" option by design.

12 · Fault Type Reference

AURA simulates eight fault types derived from actual ISS ECLSS anomaly data. Each fault injects drift into specific sensor parameters at calibrated rates.

Cabin Leak

Precursor window: 8 hours

Affected parameters: Cabin pressure ↓, N_2 ↓, O_2 partial pressure ↓, CO_2 ↓, Humidity ↓, Temperature ↓, Airflow rate ↑

MTBF: 43,800 h (5-year mean)

Remediation: Systematic hatch isolation, ultrasonic leak detection, RTV sealant or fitting replacement

SMAC trigger: Cabin pressure < 14.0 psia or drop rate > 0.05 psia/hr sustained 2h

O_2 Generator Failure

Precursor window: 24 hours

Affected parameters: O_2 output rate ↓, O_2 partial pressure ↓, O_2 purity ↓, H_2 (ppm) ↑, Cabin pressure ↓, CO_2 ↑

MTBF: 3,104 h actual (pre-flight estimate was 8,437 h — 63% worse than expected)

Remediation: Inspect/replace cell stack; purge gas bubbles from electrolysis cells; check H_2 separator membrane

SMAC trigger: O_2 partial pressure < 19.5% or H_2 > 2% in O_2 stream

O_2 Leak

Precursor window: 4 hours

Affected parameters: O_2 partial pressure ↓, Cabin pressure ↓, O_2 (mass spec) ↓, Airflow rate ↑

MTBF: 8,760 h (~annual inspection cycle for plumbing seals)

Remediation: Inspect O_2 supply fittings and isolation valve seals; close O_2 isolation valve if leak confirmed

SMAC trigger: O_2 partial pressure > 23.5% or rising trend > 0.5%/hr

CO_2 Scrubber Failure

Precursor window: 4 hours

Affected parameters: CO_2 partial pressure ↑, CO_2 (mass spec) ↑, O_2 partial pressure ↓, Temperature ↑

MTBF: 4,380 h (~6-month major service per SSP-57003)

Remediation: Replace Zeolite 13X desiccant/sorbent beds; inspect selector valves; check heater elements

SMAC trigger: CO_2 partial pressure > 0.70% (5.3 mmHg); >1.0% is crew health limit

CHX Failure (Condensing Heat Exchanger)

Precursor window: 12 hours

Affected parameters: Humidity ↑, Temperature ↑, Bacterial/fungal count ↑, Airflow rate ↓

MTBF: 17,520 h (2-year mean; hydrophilic coating degrades over 2–4 years)

Remediation: Inspect and re-apply hydrophilic coating; clean microbial growth; check condensate separator disk

SMAC trigger: Cabin RH > 70% sustained > 48h or temperature > 27°C

Water Processor Failure

Precursor window: 48 hours

Affected parameters: Water purity ↓, Production rate ↓, Bacterial/fungal count ↑, H_2O ↓

MTBF: 3,850 h actual; sieve beds replaced annually on schedule

Remediation: Replace iodinated resin sieve beds; replace multifiltration beds when TOC > 3 mg/L; inspect ion exchange beds

SMAC trigger: TOCA TOC > 3 mg/L — immediate water processing halt required

Trace Contaminant Filter Saturation

Precursor window: 72 hours

Affected parameters: NH_3 ↑↑, CH_4 ↑, CO ↑↑, H_2 (ppm) ↑

MTBF: 19,746 h (~2.25 years; no TCCS flight failures in 20+ years of ISS ops)

Remediation: Replace activated charcoal bed and LiOH bed on saturation confirmation; 90-day sampling to track loading

SMAC trigger: NH_3 > 2.6 ppm (180-day SMAC) or any VOC above SMAC-1 limits

NH_3 Coolant Leak

Precursor window: 2 hours (fast-developing)

Affected parameters: NH_3 ↑↑↑, Temperature ↑, O_2 ↓, Cabin pressure ↑

MTBF: 17,520 h (~2-year mean for major EATCS loop leak events)

Remediation: Inspect pump module fittings and quick-disconnects; replace external pump module (EPM) if leak confirmed

SMAC trigger: NH_3 > 10 ppm requires PPE; > 25 ppm is emergency evacuation threshold

13 · Sensor Parameter Reference

All 20 ECLSS parameters monitored per location, with nominal operating ranges and units from NASA-STD-3001, JSC-20584 SMAC limits, and ISS ECLSS operational data.

PARAMETER	UNIT	NOMINAL RANGE	SUBSYSTEM	SOURCE STANDARD
O_2 partial pressure	%	19.5 – 23.1	Atmosphere Revitalization	NASA-STD-3001 Vol.1 Table 6.2.1
CO_2 partial pressure	%	0.13 – 0.70	Atmosphere Revitalization	SMAC limits (5.3 mmHg caution)
Humidity	%	25 – 70	Atmosphere Revitalization	NASA-STD-3001 max 70% RH
O_2 output rate (generator)	kg/day	2.0 – 9.0	Oxygen Generation System	OGS rated 5.4 kg/day nominal
O_2 purity (generator)	%	99 – 100	Oxygen Generation System	Electrolyzer target >99%
Water purity	mg/L TOC	0.0 – 3.0	Water Recovery System	JSC-62802 potable limit <3 mg/L
Production rate (WRS)	kg/day	30 – 35	Water Recovery System	WPA design capacity
Temperature	°C	18 – 27	Temperature & Humidity Control	NASA-STD-3001 comfort range
NH_3	ppm	0 – 1	Trace Contaminant Control	180-day SMAC = 2.6 ppm; caution at 10 ppm
H_2 (%)	%	0 – 0.1	Trace Contaminant Control	OGS crossover alarm at 2%; LEL at 4%
H_2 (ppm)	ppm	0 – 10	Trace Contaminant Control	MCA dissolved/trace H_2 monitoring
CO	ppm	0 – 5	Trace Contaminant Control	180-day SMAC = 10 ppm; 7-day SMAC = 15 ppm
Airflow rate	m/s	0.1 – 1.0	Air Circulation / Ventilation	IMV/CCAA target 0.1–1.0 m/s
Cabin pressure	psia	14.0 – 14.9	Pressure Control	ISS ops target 14.7 psia
Bacterial/fungal count	CFU/mL	0 – 50	Microbial Monitoring	ISS microbial limit 50 CFU/mL
N_2	%	75 – 80	Mass Spectrometer Module	Balance gas fraction
O_2 (mass spec)	%	19.5 – 23.5	Mass Spectrometer Module	MCA O_2 channel

PARAMETER	UNIT	NOMINAL RANGE	SUBSYSTEM	SOURCE STANDARD
CO_2 (mass spec)	%	0.13 – 0.70	Mass Spectrometer Module	MCA CO_2 channel
CH_4	ppm	0 – 10	Mass Spectrometer Module	Sabatier byproduct; SMAC-1 = 5300 ppm
H_2O (mass spec)	% RH	20 – 60	Mass Spectrometer Module	MCA humidity channel

Correlated noise: Parameters are not generated independently. AURA uses a Cholesky-decomposed correlation matrix (e.g., $O_2 \leftrightarrow CO_2$: $r = -0.82$, Temperature $\leftrightarrow$ Humidity: $r = +0.65$) to produce physically realistic cross-parameter noise. This means anomaly detection must handle correlated sensor drift, matching real-world ECLSS behavior.

14 · ML / AI Systems Reference

14.1 Isolation Forest (Anomaly Detection)

The Isolation Forest (IF) is trained on nominal sensor data only. It operates on the complete 20-parameter sensor vector for a given location.

- **Output:** Binary label — `+1` (nominal) or `-1` (anomalous)
- **Speed:** Sub-millisecond inference; runs every tick
- **Principle:** Anomalous points are easier to isolate by random feature splits (fewer splits needed)
- **Training:** Fit on clean nominal data; model file: `models/isolationForestModel.joblib`
- **Role in pipeline:** First-stage filter. Does not identify fault type — only flags statistically unusual readings

14.2 Random Forest Classifier (Fault Classification)

The Random Forest (RF) runs on every tick regardless of the IF label, providing fault classification across all 8 fault types plus a "nominal" class.

- **Output:** Predicted fault class + confidence probability (0–1)
- **Bypass threshold:** When RF confidence ≥ 0.85, the DQN uses the RF prediction directly rather than the raw state vector
- **Latch logic:** 3+ consecutive ticks at ≥ 95% confidence for the same fault class triggers the fault latch popup
- **Training:** Fit on labeled fault-drift data; model file: `models/randomForestModel.joblib`

14.3 LSTM Predictor (Failure Probability + RUL)

The LSTM (Long Short-Term Memory) network with multi-head attention processes a rolling window of recent sensor readings to produce two outputs:

Failure Probability

Scalar 0–1. The probability of imminent subsystem failure based on recent trends. Maps to NOMINAL / ELEVATED / WARNING / CRITICAL status words displayed on Dashboard cards.

Remaining Useful Life (RUL)

Estimated hours until failure, derived from FAULT_PRECURSOR_HOURS calibration data. Displayed in hours on Dashboard cards. Only meaningful when failure_prob > 0.2.

- **Rolling buffer:** The LSTM maintains a per-location rolling buffer of the last N sensor readings. Predictions are not issued until the buffer is sufficiently filled.
- **Architecture:** LSTM layers + multi-head attention + fully connected regression head
- **Model file:** `models/lstm_predictor.pt`

> **Cold start:** After clearing LSTM buffers or starting fresh, the LSTM will show no predictions until the rolling buffer accumulates enough readings (typically 30–60 ticks). This is expected behavior.

14.4 DQN Recommender (Remediation Actions)

The Deep Q-Network (DQN) maps the current system state to one of 11 possible remediation actions. It is trained via reinforcement learning with a reward structure that prioritizes restoring nominal conditions.

- **State vector:** Current sensor readings, IF label, RF classification, LSTM failure_prob, and RUL for a location
- **Output:** Action index + confidence. The action name is displayed on each Dashboard card and in the Alerts view.
- **RF bypass:** When RF confidence ≥ DQN bypass threshold, the DQN receives the RF-identified fault as part of its state, improving action specificity
- **Fallback:** If the DQN model is not loaded, a rule-based fallback maps RF fault class directly to the corresponding remediation action
- **Model file:** `models/dqn_recommender.pt`

ACTION	CORRESPONDING FAULT
No Action Needed	(nominal state)
Use Sealant to Close Leak	Cabin Leak
Remove gas bubbles from O_2 generator	O_2 Generator Failure
Close Oxygen Isolation Valve	O_2 Leak
Remove and Replace Air Selector Valves	CO_2 Scrubber Failure
Replace CCAA Heat Exchanger	CHX Failure
Replace MF beds or Ion Exchange Beds	Water Processor Failure
Replace Active Charcoal Bed	Trace Contaminant Filter Saturation
Close Hatch and replace external pump module	NH_3 Coolant Leak
Use Redundant Sensors on Pump	(sensor validation)
Reopen Reference Gas Valve	(valve state correction)

15 · Troubleshooting

SYMPTOM	LIKELY CAUSE	RESOLUTION
Server fails to start: `ModuleNotFoundError`	Package not installed in `.venv`	Delete `run/linux/.req_hash` (or `run/windows/.req_hash`) and re-run the start script. It will reinstall all packages.
Settings API returns 422 Unprocessable Entity	Route handler missing `: Request` type annotation	All settings route handlers must have `request: Request` typed parameters. Verify main.py is up to date.
Sensor Detail shows 404 for Bacterial/fungal count	Parameter name contains `/` which Starlette decodes in path segments	Parameter must be passed as a query parameter (`?parameter=Bacterial%2Ffungal+count`), not a path segment. Verify the API endpoint uses `?parameter=`.
Settings forms are blank after login	GET /api/settings returning 422 (see above)	Fix request type annotations; form pre-population is automatic once the endpoint returns 200.
AI Analyst shows "No AI backend available"	Ollama not running and no Groq key configured	Start Ollama (`ollama serve`) and pull a model (`ollama pull mistral`), OR configure a Groq API key in Settings → Integrations.
LSTM shows no predictions (blank)	Rolling buffer has not filled yet (cold start)	Wait for 30–60 ticks after server start or after clearing LSTM buffers. Normal behavior.
Trends page shows almost all sensors as Advisory	Thresholds too sensitive for current noise level	Increase MK τ advisory threshold (e.g., 0.45), slope gate (e.g., 0.08), and z-score threshold (e.g., 4.0) in Settings → Trend Detection.
WebSocket indicator stays red (disconnected)	Server not running, port blocked, or browser restriction	Confirm server is running at port 8000. Check firewall. Browser auto-reconnects every 3 seconds.
Fault injection has no visible effect	Fault injection master switch is disabled	Settings → Data Generation → Fault injection enabled must be checked. Also confirm the fault was successfully injected (check Active Faults table in Settings → Faults).
CSV export downloads empty file	Database has no sensor data	Wait for at least one data generation tick (the background loop runs every 1 second). Check server logs for generation loop errors.
Admin password forgotten	bcrypt hash in settings.json	Delete `settings.json`. On next server restart, `settings_manager.py` will re-create it with the default password (`admin`). Note: this also resets all other settings to defaults.
ML models not loading (red ✗ in Settings → ML Models)	Model files missing from `models/` directory	Run the training scripts in `scripts/` (e.g., `python scripts/train_isolation_forest.py`). Models are not included in the repository and must be trained locally.

SYMPTOM	LIKELY CAUSE	RESOLUTION
Groq API key: "Test Connection" fails	Invalid key, expired free tier, or network issue	Regenerate the key at console.groq.com. Re-save via Settings → Integrations. Check that the server has internet access on port 443.

16 · Technical Reference

File Structure

```
Final-AURA/
├── main.py              — FastAPI app, all routes, background loop, auth
├── database.py          — SQLite persistence layer (WAL mode)
├── data_generator.py    — Physics-correlated ECLSS sensor simulator
├── ml_pipeline.py       — Isolation Forest + Random Forest inference
├── lstm_predictor.py    — LSTM multi-head attention predictor
├── dqn_recommender.py   — Deep Q-Network remediation recommender
├── trend_detector.py    — Mann-Kendall / CUSUM / z-score analysis
├── ai_analyst.py        — Ollama / Groq AI Analyst backend
├── settings_manager.py  — Persistent settings, bcrypt auth, Fernet encryption
├── constants.py         — All static configuration (locations, faults, ranges)
├── requirements.txt     — Python dependencies
├── settings.json        — Runtime settings (auto-created; do not commit)
├── .fernet.key          — Encryption key (auto-created; never commit)
├── data/
│   └── aura.db          — SQLite database (WAL mode)
├── models/
│   ├── isolationForestModel.joblib
│   ├── randomForestModel.joblib
│   ├── lstm_predictor.pt
│   └── dqn_recommender.pt
├── scripts/
│   ├── train_isolation_forest.py
│   ├── train_random_forest.py
│   ├── train_lstm.py
│   └── train_dqn.py
├── static/
│   ├── index.html
│   ├── css/style.css
│   └── js/
│       ├── app.js       — SPA controller (~2000 lines)
│       └── twin.js      — Three.js Digital Twin module
├── run/
│   ├── linux/start.sh
│   └── windows/start.bat
└── manual/
    └── aura_manual.pdf  — This document
```

REST API Endpoints (Key)

METHOD	PATH	DESCRIPTION
GET	`/api/config`	Static configuration: locations, fault types, parameter ranges, ML status
GET	`/api/locations`	Current anomaly state and active fault for all locations
GET	`/api/history`	Parameter history: `?parameter=X&n=100&location=Y`
POST	`/api/faults/inject`	Inject a fault: `{"location": "...", "fault": "..."}`

METHOD	PATH	DESCRIPTION
DELETE	`/api/faults`	Clear all injected faults
DELETE	`/api/faults/{location}`	Clear fault for a specific location
GET	`/api/alerts`	Alert history with optional filters
POST	`/api/auth/login`	Authenticate: returns JWT token
GET	`/api/settings`	All settings (auth required)
PATCH	`/api/settings/alerts`	Update alert thresholds (auth)
PATCH	`/api/settings/trends`	Update trend detection thresholds (auth)
GET	`/api/settings/data/export/csv`	Stream CSV export (auth); X-Export-Max-Id header
DELETE	`/api/settings/data/exported`	Clear exported rows by watermark (auth)
GET	`/api/trends/{location}`	Run trend analysis for a location
GET	`/api/maintenance`	MTBF and calibration schedule data
WS	`/ws/live`	WebSocket live tick feed

Database Schema

```
faults         (id, fault_name UNIQUE)
locations      (id, location_name UNIQUE, current_fault_id → faults.id)
generated_data (id, location_id → locations.id, data TEXT/JSON, timestamp)
anomaly_labels (id, data_row_id → generated_data.id UNIQUE,
                isolation_forest_label, random_forest_classification)
alerts         (id, location_name, timestamp, severity,
                fault_type, top_probability, sensor_data, acknowledged)
```

WebSocket Tick Message Format

```
{
  "type": "tick",
  "location": "US Lab",
  "timestamp": "2025-01-01T12:00:00",
  "data": { "O2 partial pressure": 21.2, "CO2 partial pressure": 0.38, ... },
  "is_anomalous": false,
  "rf_classification": { "prediction": "nominal", "confidence": 0.92 },
  "lstm": { "failure_prob": 0.05, "rul_hours": 240.0 },
  "dqn": { "action": "No Action Needed", "action_index": 0, "confidence": 0.87 },
  "latched_fault": null
}
```

Settings Persistence

Settings are stored in `settings.json` at the project root. On server startup, `settings_manager.py` loads this file (or creates it from defaults) and applies all values to the running module variables via `apply_to_main()`, `apply_to_dqn()`, and `apply_to_trend_detector()`. No restart is required for threshold changes to take

effect. The password bcrypt hash and encrypted Groq key are stored in this file; the Fernet encryption key is in a separate `.fernet.key` file.